

Product Requirements Document (PRD)

Product Name

ScotJob Pulse AI

Real-Time Scotland Job Monitoring + Telegram Alert System

Version: MVP v1

Owner: Nithin Peddapelly

Status: Draft

1. Product Vision

Build a personal job-monitoring platform that continuously monitors Amazon Jobs and future job sources, detects new jobs before public Telegram repost channels, filters only relevant Scotland jobs, and immediately sends alerts to Telegram.

Telegram acts only as a **delivery mechanism**.

The scraper and monitoring engine perform the actual work.

2. Problem Statement

Current methods create delays:

- Telegram channels repost jobs after detection delay
- Users manually refresh websites
- Warehouse jobs disappear quickly
- Users receive irrelevant jobs from all over the UK
- Fast-moving listings often fill before users discover them

Goal:

Enable the user to receive Scotland job alerts before public channels.

3. Goals

Primary Goals

- Detect jobs faster than Telegram repost channels
 - Only send Scotland jobs
 - Prevent duplicate alerts
 - Filter based on user preferences
 - Support near real-time monitoring
-

Secondary Goals

- Expand to more job sources
 - Add AI ranking
 - Add dashboard and analytics
-

4. User Persona

Primary User

International student / part-time worker

Needs:

- Fast warehouse jobs
 - Part-time jobs
 - Scotland locations only
 - Immediate notifications
 - Low manual effort
-

5. User Stories

Detection

As a user

I want new jobs detected immediately

So I can apply before positions disappear.

Scotland filtering

As a user

I only want Scotland jobs

So I avoid irrelevant UK listings.

Preferences

As a user

I want filtering by keywords and work type

So I only see jobs I care about.

Notifications

As a user

I want Telegram alerts instantly

So I can apply immediately.

6. Functional Requirements

FR-1 Job Monitoring Engine

Initial source:

- Amazon Jobs

Future sources:

- Indeed
- LinkedIn
- Reed
- Company career pages
- Tesco
- NHS
- Sainsbury

Requirements:

- Poll at configurable intervals
- Respect rate limits
- Store monitoring logs
- Retry on failures

Input:

sourceURL

region

pollingInterval

Output:

job listings

FR-2 Detection Engine

Requirements:

- Fetch jobs
- Extract metadata
- Compare against database
- Detect unseen jobs only

Logic:

```
fetchJobs()
```

```
for(job of jobs){
```

```
  if(job.id not in database){
```

```
    saveJob()
```

```
    sendAlert()
```

```
  }
```

```
}
```

FR-3 Scotland Filter Layer

The system must only send Scotland jobs.

Primary detection:

```
location.includes("scotland")
```

Secondary fallback:

City matching.

Supported Scotland cities/areas:

```
const scotlandLocations=[
```

```
  "bathgate",
```

```
  "dunfermline",
```

```
  "motherwell",
```

```
  "baillieston",
```

```
"aberdeen",  
"edinburgh",  
"livingston",  
"glasgow",  
"fife",  
"west lothian",  
"musselburgh",  
"coatbridge",  
"airdrie",  
"cumbernauld",  
"broxburn",  
"scotland"
```

```
]
```

Filtering logic:

```
function isScotlandJob(location){
```

```
location=location.toLowerCase()
```

```
return location.includes("scotland")
```

```
||
```

```
scotlandLocations.some(  
city=>location.includes(city)  
)
```

```
}
```

Pipeline:

Amazon Jobs

↓

Scraper

↓

Extract Location



Scotland Filter



Deduplication



Database



Telegram Notification

FR-4 Filtering System

Allow filtering by:

Job title keywords

Examples:

Warehouse

Kitchen Porter

Software Tester

Barista

Chef

Location

Examples:

Edinburgh

Bathgate

Dunfermline

Livingston

Salary

Example:

Minimum £13/hr

Work type

Options:

- Full-time
- Part-time
- Internship
- Seasonal
- Contract

FR-5 Deduplication

Prevent:

- Duplicate notifications
- Duplicate processing
- Repeated jobs

Logic:

```
if(jobID exists){
```

```
    ignore()
```

```
}else{
```

```
    save()
```

```
    notify()
```

```
}
```

FR-6 Telegram Notification System

Requirements:

- Use Telegram Bot API
- Never scrape Telegram
- BOT_TOKEN stored in environment variables
- Send alerts to personal DM

Notification example:

 New Scotland Job Found

 Bathgate, Scotland

 Warehouse Operative

 Seasonal | Part-time

 £14.30/hr

 16 hrs/week

Apply:

<https://job-link>

7. Architecture

Amazon Jobs Website

Indeed

LinkedIn

Career Pages



+-----+

| Monitoring Service |

+-----+



+-----+

| Extraction Engine |

+-----+




```
+-----+
| Scotland Filter |
+-----+
```



```
+-----+
| Filter Engine   |
+-----+
```



```
+-----+
| Deduplication Logic |
+-----+
```



```
+-----+
| Database        |
+-----+
```



```
+-----+
| Telegram Service |
+-----+
```



Telegram Notification

8. Folder Structure

scotjob-pulse/

|— src/

```
|
|
|— config/
|   |— telegram.js
|   |— database.js
|
|— scrapers/
|   |— amazonScraper.js
|   |— indeedScraper.js
|   |— linkedInScraper.js
|
|— services/
|   |— monitorService.js
|   |— filterService.js
|   |— dedupeService.js
|   |— notificationService.js
|
|— database/
|   |— schema.prisma
|
|— routes/
|   |— jobs.js
|
|— utils/
|   |— logger.js
|
|— .env
|— package.json
|— server.js
```

9. Database Schema

Jobs Table

Field	Type
-------	------

id	UUID
sourceJobID	String
title	String
location	String
salary	String
workType	String
source	String
link	String
detectedAt	Timestamp

Users Table

Field	Type
id	UUID
telegramID	String
keywords	String
preferredLocation	String
minSalary	Number

10. Suggested Tech Stack

Backend:

- Node.js

- Express

Libraries:

axios

cheerio

playwright

node-cron

dotenv

node-telegram-bot-api

prisma

Database:

Development:

SQLite

Production:

PostgreSQL

Caching:

Redis

Deployment:

[Render](#)

[Railway](#)

[Docker](#)

11. MVP Scope

Included:

- ✓ Amazon monitoring
- ✓ Scotland-only filtering
- ✓ Telegram alerts
- ✓ Deduplication
- ✓ Database storage
- ✓ Keyword filtering

Excluded:

- ✗ Dashboard
- ✗ AI scoring

12. Estimated Timeline

Phase	Estimate
Telegram setup	1 day
Scraper engine	2 days
Scotland filter	1 day
Database integration	1 day
Notifications	1 day
Testing	2 days
Deployment	1 day

Estimated MVP:

7–10 days